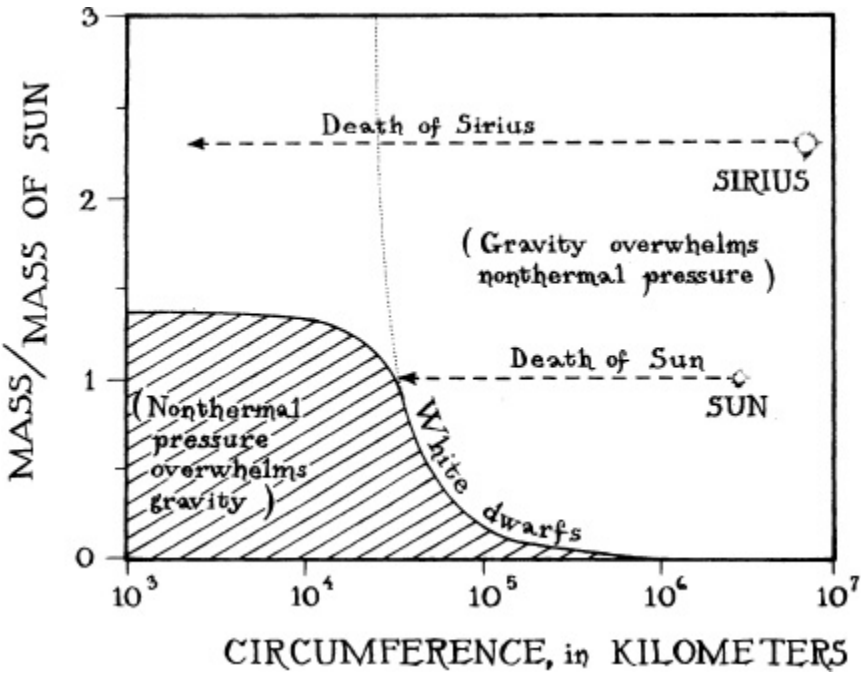



**4.4 When a normal star such as the Sun or Sirius (not Sirius B) starts to cool off, it must shrink, moving leftward in this diagram of mass versus circumference. The shrinkage of the Sun will stop when it reaches the edge of the shaded region (the white-dwarf curve). There degeneracy pressure balances gravity's squeeze. The shrinkage of Sirius, by contrast, cannot be so stopped because it never reaches the edge of the shaded region. See Box 4.2 for a different depiction of these conclusions. If, as Eddington claimed, white-dwarf matter's resistance to compression were always 5/3, that is, if relativity did not reduce it to 4/3 at high densities, then the graph of mass versus circumference would have the form of the faint dotted curve, and the shrinkage of Sirius would stop there.**

After the meeting, one fellow after another came up to Chandrasekhar to offer condolences. "I feel it in my bones that Eddington is right," Milne told him.

The next day, Chandrasekhar began appealing to his physicist friends for help. To Leon Rosenfeld in 'Copenhagen, he wrote, "If Eddington is right, my last four months' work all goes in the fire. Could Eddington be right? I should very much like to know Bohr's opinion." (Niels Bohr was one of the fathers of quantum mechanics and the most respected physicist of the 1930s.) Rosenfeld replied two days later, with assurances that he and Bohr both were convinced that Eddington was wrong and Chandrasekhar right. "I may say that your letter was *some* surprise for me," he wrote, "for nobody had ever dreamt of questioning the equations [that you used to derive the 4/3